Chia-Yuan Chiang

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Education

Yuan Ze University — Taoyuan, Taiwan
B.S. in Electrical Engineering — *Sep. 2024 – Sep. 2027 (Expected)*

Ming Chi University of Technology — New Taipei, Taiwan
B.S. in Electrical Engineering (Transferred) — *Sep. 2023 – Jun. 2024*

Nangang Senior High School — Taipei, Taiwan
Electronics Technology — *Sep. 2020 – Jun. 2023*

Experience

Undergraduate Researcher — May 2025 – Present
Prof. Shu-Yen Lin's Lab, Yuan Ze University — *Taoyuan, Taiwan*

- Led an early-warning model for ICU Ventilator-Associated Pneumonia (VAP); designed a stay-level 5-fold cross-validation scheme to eliminate patient-level data leakage present in the public benchmark.
- Applied Integrated Gradients feature attribution to reduce the model to 4 non-invasive features; the resulting LSTM reached 0.98–0.99 AUROC at a 72-hour prediction horizon.
- Manuscript submitted to IEEE GCCE 2026.

Engineering Advisor & Systems Integrator — 2023 – Present
FRC Team 7645 — *Taipei, Taiwan*

- Own robot system architecture: multi-sensor fusion, PID closed-loop motor control tuning, and software/hardware interface definitions.
- Introduced LLM-assisted tooling that cut iterative firmware debugging time by roughly one week; trained junior members to debug from root-cause analysis instead of trial-and-error.
- Rebuilt the team website, reducing repetitive PR/outreach inquiries.

Competitor — Jun. 2020 – Sep. 2023
Nangang Senior High School – NK-MTC 7645 — *Taipei, Taiwan*

- Generalist across LabVIEW/Arduino firmware, FRC electrical wiring, AutoCAD modeling, and 3D-printed mechanical parts.
- Competed in 3 FRC regional events (2020 CIT 5G Regional, 2022 Hon Hai Taiwan Regional) and represented the school at the National Skills Competition.

Projects

ICU VAP Early-Warning Model | *PyTorch, LSTM, SHAP* — 2025 – Present

- Single-center pilot (Far Eastern Memorial Hospital ICU, n=109); designed stay-level 5-fold stratified CV that exposed patient-level data leakage in the PREDICT 2025 benchmark (AUROC dropped from an inflated 0.99 to a real 0.58).
- Used Integrated Gradients to drop a collinear redundant feature ($mv\_total \equiv Vt \times RR$) and reduced the model to 4 non-invasive features; final LSTM reached 0.99 AUROC at a 24-hour horizon, 0.98–0.99 across 6–72h.
- Outperformed LR/RF/SVM baselines at every prediction window under the same setup; manuscript submitted to IEEE GCCE 2026.

PCB Defect Detection | *Azure Custom Vision, Python* — 2025

- Trained an object-detection model to localize PCB defects across 3 board types and 3 defect classes; first attempt on full-resolution images scored only 9.6% mAP because each defect box (~60px) vanished after Custom Vision's resize.
- Fixed it with 600px defect-centered tiling, raising defect bounding-box area share from ~2% to ~12% of each tile; mAP went from 9.6% to 81.9%.
- Validated on held-out board types (06/07) unseen during training; reached 100% precision, 69.7% recall. Exported TensorFlow model.pb and ran inference in Colab with the official anchor/NMS decoder.

Open-Source QMK x STM32 Numpad | *C, ChibiOS HAL, EasyEDA* — 2026

- Designed a 17-key PCB in EasyEDA Pro, hand-soldered the board, and flashed STM32F072 via DFU.
- Reverse-engineered ChibiOS HAL config (chconf/halconf/mcuconf) to resolve build conflicts and timing issues between the open-source PCB and QMK firmware; reached stable USB HID enumeration with ¡5ms key latency.

- Got RGB Matrix working with a brightness cap to prevent LED over-current, plus live VIA/VIAL remapping without recompiling; fixed a top/bottom-layer footprint mirroring bug in the PCB design.

Swerve Drive Chassis (FRC) | *LabVIEW, CNC, PID Control* 2020 – 2023

- Designed and machined 3 generations of a swerve-drive chassis from scratch: kinematics, gear train, CNC/wire-EDM fabrication, and LabVIEW PID control.
- Gen-3 cut weight via 3D-printed gears, modularized drive/steer motor mounts (NEO or Falcon 500 interchangeable), and redesigned wheels for better grip; won Innovation in Control Award at 2022 FRC Taiwan Hon Hai Regional.

Job Radar | *Python, Playwright, GitHub Actions, Cloudflare Pages* 2026 – Present

- Built a personal job-listing aggregator scraping 104, Cake, and LinkedIn (~1,300 listings/day) via a daily GitHub Actions cron; reverse-engineered 104's internal SPA API after the legacy API was deprecated.
- Designed a config-driven scoring engine (role fit, location commute, experience threshold) and a static-site dashboard build pipeline with no backend or database.

AWS x BitoPro Hackathon – AI Fraud Detection 2026

- Built an account risk-scoring dashboard (AUC 83.2%) with 5-fold CV and decision-threshold tuning (optimal 0.24) over 12,753 user accounts.

Technical Skills

Languages & Firmware: C/C++ (STM32/Cortex-M), Python, QMK/ChibiOS HAL, LabVIEW, Arduino/ESP32, UART/SPI/I2C
Machine Learning: PyTorch, TensorFlow/Keras, LSTM/Time-Series, Feature Engineering, Signal Processing, SHAP/XAI, TinyML
EDA / Hardware: Altium Designer, KiCAD, OrCAD, EasyEDA Pro
Manufacturing: CNC Milling, Mastercam, 3D Printing, Laser Cutting, Welding, AutoCAD, Fusion 360, Autodesk Inventor

Awards & Certifications

2022 FRC Taiwan Hon Hai Regional – Finalist Alliance & Innovation in Control Award · 52nd National Skills Competition – Team Project Category Representative (2022) · 2020 FRC CIT 5G Regional – Excellence in Engineering (Industrial Design) Award